LECTURE NOTES.

DIFFERENCE BETWEEN TURRET LATHE & ENGINE LATHE.

The Main difference between the two Machines is that the turret lathe is adapted to Quantity production work, whereas the engine lathe is primarily used for miscellaneous jobbing tool room, or Single operation work. The essential features of a turret lathe that Makes it a Quantity production M/c are these:

(i) Tools May be permanently Set up in the turret in the proper Sequence of their use.
(ii) Each Station is provided with a feed Stop or feed trip So that each cut of a tool is the Same as the previous cut.
(iii) Multicut Can be taken from the same Station at the Same time, Such as two or more turning and/boring cuts.
(iv) Combines cuts Can be Made. That is tools on the Slide Can be used at the time that tool on the turret are cutting.
(v). Extreme rigidity in the holding of work and tools is built into the Machine to permit Multiple and and Combined cuts.
(vi) They May be fitted with various attachments, Such as for proper turning, thread chasing and duplicating and Can be Controlled.

Difference B/W capstan lathe and Turret lathe.

The Main difference b/w the capstan and Turret lathe is that on the capstan lathe, the turret is carried on and drives its Movement from an auxilliary Slide clamped to the top of the bed and adjustable along the Same for varying its distance from the Machine nose. Turret of the turret lathe is mounted on a Saddle which Slides directly on The bed in the Same way to a lathe saddle.

AIR OPERATED CHUCKS.

When a compressed air supply is available as it is now in most workshops, chucks may be fitted which are operated by this medium. This feature reduces chucking time, relieves the operator of manual fatigue and if the air valves are foot operated leaves him with both hands free to manipulate the work during chucking. These chucks are operated with air at a pressure of about 6 bars.

SOFT JAWS.

Many components after their first setting have to be rechucked for subsequent operations. Often the work has to be held on previously machined surfaces and it is important that others surfaces and bored holes are true with the surfaces held by the chucks.

Then it is common practice to replace the hardened jaws by soft ones and, after tightening the chuck at the required diameter, clean up the clamping surfaces of the jaws with a boring tool. This ensures concentricity of work, and the soft steel of the jaws avoids damage when holding previously machined surfaces.

Difference Between Shaping and Planing Machines:

When the shaping and planing machines are compared in terms of construction, operations and use, the following differences may be seen:

(i) The planner is especially adopted to large work; shaper can ~~do~~ only small work.

(ii) On the planner, the work is moved against a stationary tool. On the shaper, the tool moves across the work which is stationary

(iii) On the planer, the tool is fed into the work, on the shaper the work is usually fed across the tool.

(iv) Most planers differ from shapers in that they

approach more constant velocity cutting speed.
(v) The drive on the planer ~~ray~~ table is either by gears or by hydraulic means. The shaper ram can also be driven in this manner, but many times a quick return link mechanism is used.

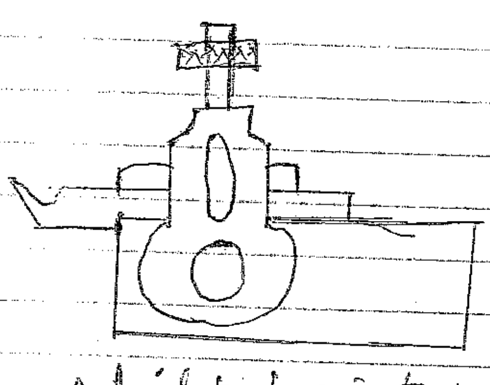
BORING.

It is the enlarging of a hole that has already been drilled or cored. Principally, it is an operation of trueing a hole that has been drilled previously with a simple point lathe type tool.

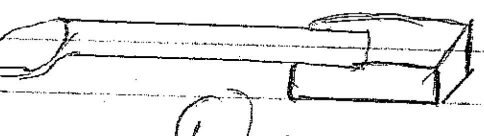
TYPES OF BORING MACHINES.

There are two main types of boring machines, viz:
Vertical Boring Mill: The vertical boring mill has a horizontal circular worktable and is adapted to facing and vertical turning as well as boring work.
Horizontal Boring Machine: This differs from the vertical mill ~~th~~ in that the work is stationary and the tool is revolved; is adopted to the boring of horizontal holes. It has horizontal spindle for holding the tool. A worktable having longitudinal and cross movements is supported on ways on the bed of the machines. In some cases, the table is capable of being swiveled to permit indexing the work and boring holes at desired angles.

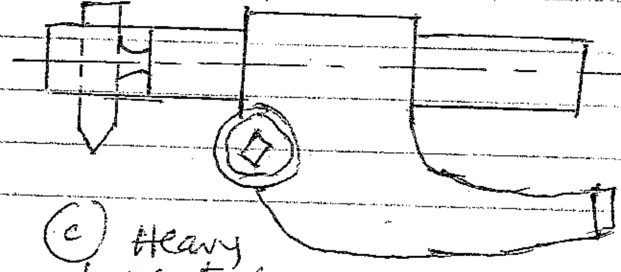
BORING TOOLS.



(a) A light boring tool with band shank



(b) forged boring tool



(c) Heavy boring tool

ADAITIVES USE IN PLASTICS.

(i) <u>Plasticisers</u>; These are particularly added to P.V.C to give greater flexibility and make it easier to form!

(ii) Stabilisers: These are added, again particularly to P.V.C. to prevent decomposition of the polymer at temperatures encountered during normal processing.

(iii) Antioxidants: These are added to prevent break-down in the presence of oxygen, which most polymers are subject to at the elevated temperatures necessary for processing and at atmospheric temperatures over a period of time.

(iv) Lubricants; widely used to facilitate the processing of a variety of polymers by reducing the forces between molecules and by reducing adhesion of the polymers to hot metal surfaces during processing.

<u>~~[illegible]~~ FLOOR - TO - FLOW TIME:</u>

MEP 404 / MEG 422
LECTURES
METAL CUTTING

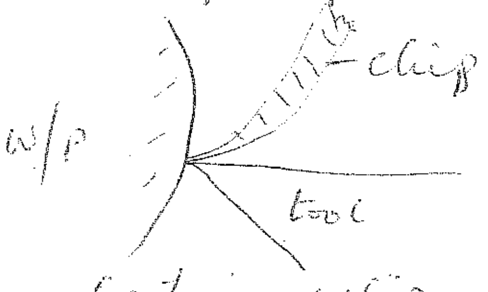
This is the process of removing chips from a rough block of Material in order to obtain the desired Shapes, Size and Surface.

The most important Variable in Metal Cutting are:-

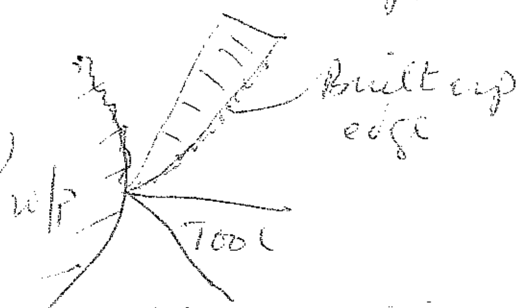
(i) The properties of the work Material
(ii) The properties and geometry of the cutting tool
(iii) The interaction b/w the tool and the work during cutting.

CHIP FORMATION.

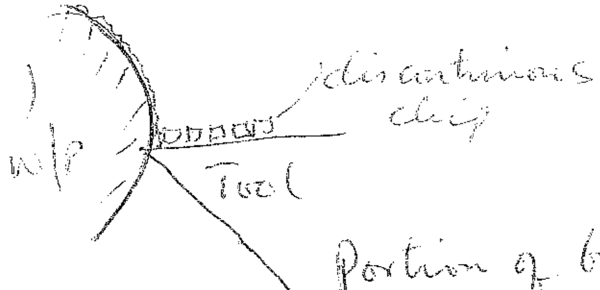
Basically there are three types of chips formed during Metal cutting. These include Continuous chip, Continuous chip with Build up edges and discontinuous chip.

i) 

Continuous chip

(ii) 

Continuous chip with up edge.

(iii) 

Portion of built up edges adhering to chip and work piece refer to diagram (ii) (1) above his is similar to the above but differs as a result of small particles from the work piece being built up and welded to the tool face under the heavy pressure and the heat generated at the top of the tool face.

The discontinous chips leaves the tool as small segments of metal and results from cutting brittle metal such as cast iron.

FACTORS RESPONSIBLE FOR THE FORMATION OF CHIPS.

(i) CONTINOUS CHIP.

a) high cutting speed
b) Large rake angle
c) Small depth of cut
d) efficient cutting fluid.

ii) DISCONTINOUS CHIP.

a) Low cutting speed
b) Small rake angle
c) Large depth of cut
d) No or inefficient cutting fluid.

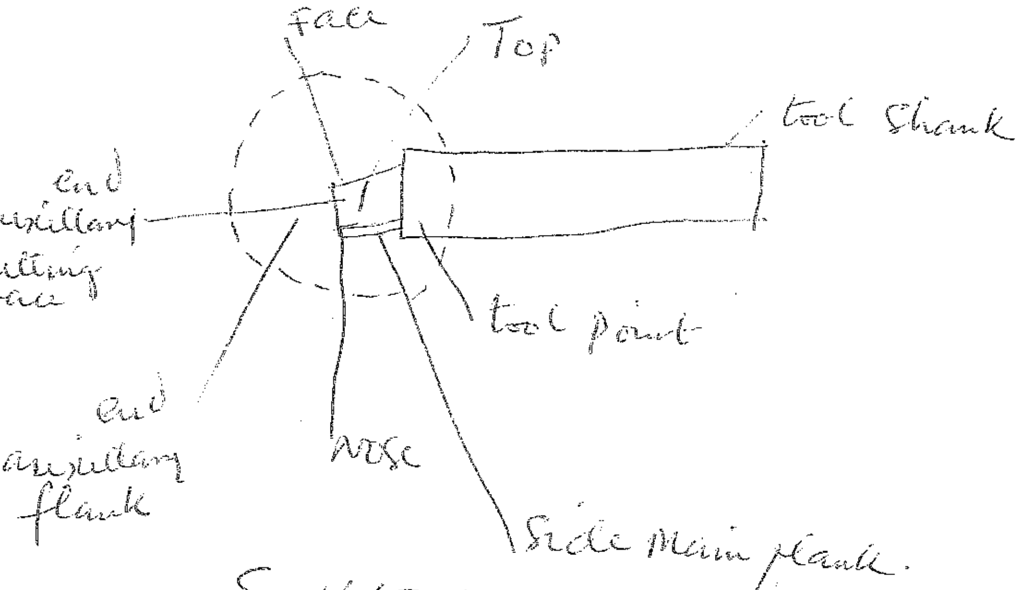
iii) CONTINUOUS CHIP WITH (Built-up edges)

a) low cutting speed
b) Large depth of cut
c) No cutting fluid
d) Poor surface finish.

WHY BUILT up-EDGE ON A CUTTING IS UNDESIRABLE

(i) It is a Source of Vibration and this May Cause tool breakage.

(ii) Poor Surface finish is usually the effect of Built-up edge.



SINGLE POINT CUTTING TOOL

CUTTING TOOL MATERIALS

(i) Tungsten Carbide: This is a very hard and brittle alloy and is used to Machine Such Materials as grey cast Iron and Cast bronze. They have low tensile Strength and form discontinuous chips. This tool Material is not Suitable for Machining ductile Materials

(ii) High Speed Steels: This Consist of an alloy of Iron and Carbon with different amount of other alloying elements Such as tungsten, Chromium, Vanadium and Cobalt. When hardened, they are brittle and the cutting edge will chip on impact or with rough handling. They have high resistance to abrassion but not tough enough to withstand shock loads. They can cut at high speed and maintain hardness value at high temperatures during cutting.

iii) Cemented carbide: They are produced by every powder metallurgy principle. The mixture of powder of various amount of hard particles and binding metal. The hard particles gives the material its hardness and abrassive resistance while the binding metal provide the toughness.

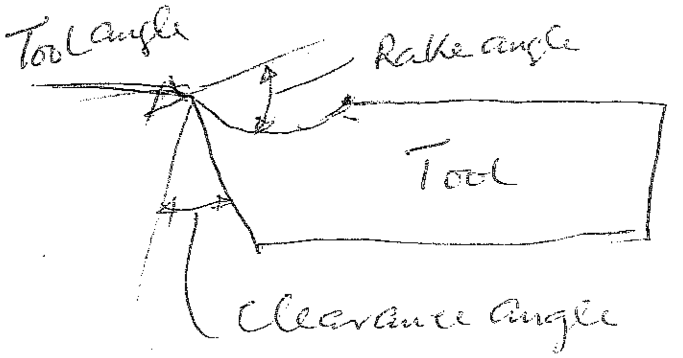
iv) Stellite: This is a cobalt-chromium-tungsten alloy containing no iron. It can not be rolled or forged and is shaped by casting from which it derives its cutting properties and hardness. No heat treatment is required after the casting. It is as hard as high speed steel and has higher red-hardness, retaining its hardness at higher temperatures.

CUTTING TOOL ANGLES.

i) Clearance angle: This is a clearance b/w the work piece and the tool to avoid rubbing of the tool on the work piece. If such clearance is not provided, friction and heat resulting from the rubbing action will cause tool to wear out quickly. It is kept as a small as possible (5° to 7°). If the clearance angle is made too large, the tool point will be weakened and the tool may "dig in" thus producing a poor surface finish.

(ii) Rake angle: This is the angle that the cutting face of a tool makes with a line perpendicular to the work surface. Large rake angle are good for cutting ductile materials. It makes cutting easier and less effort is expended. However, too large rake angle makes the tool point considerably weakened. Clearly a compromise must be reached.

(iii) Tool angle: The tool angle size determines the strength of the tool point, small tool angles makes tool point weak and large tool angle makes the tool stronger and robust.



TOOL LIFE: This is time between which tool is put to use and the time it can no longer perform its desired function effectively and required either reshaping or complete replacement.

RED HARDNESS: This is ability of cutting tool to retain its hardness at high cutting temperatures

FACTORS THAT AFFECT TOOL LIFE

(i) Cutting speed: As cutting increases, temperature also increase (rise). The heat generated is more [illegible] lathe tool [illegible] on the tool than on work and the hardness of tool materials change. This weakens the tool and hence reduces tool life.

(ii) Feed and depth of cut: Tool life is greatly influenced by the feed rate and the depth of cut (chip thickness), low feed rate and appropriate depth of cut prolong tool life compared to higher feed rate and high depth of cut.

(iii) Tool geometry: Tool life is also affected by tool geometry, a tool with large rake angle becomes weak as a large rake angle reduces the tool cross section and the amount of metal to absorb heat.

(iv) Tool Material: Physical and chemical properties of tool materials affect tool life by affecting stability and rate of wear of tool.

(v) Cutting fluid: This reduces the coefficient of friction at the tool interface and increases tool life.

CUTTING TOOL DYNAMOMETER.

This is an instrument designed to detect small deflections of the cutting tool when it is subjected to forces. Such as the cutting forces and feed force on the cutting tool. It is used to measure cutting tool forces.

SIMPLE CALCULATIONS ON LATHE TOOL DYNAMOMETER.

QUESTION

During a test using a lathe tool dynamometer the following results were obtained

(i) Spindle speed — 650 rev/min
(ii) Depth of cut — 10mm
(iii) Blank diameter of work — 100mm
(iv) Feed — 0.5 mm/rev
(v) Vertical cutting force — 1700 N
(vi) Feed force — 500N

determine the total power consumed at the tool point.

SOLUTIONS.

GIVEN.

- Spindle speed — 650 rev/min
- depth of cut — 10 mm
- Blank dia. of w/k — 100mm.
- Feed — 0.5mm/rev
- Vertical cutting force — 1700N
- Feed force — 500N.

(i) Power due to cutting $= \dfrac{F_c \times N \times DN}{1000 \times 60} = \dfrac{1700 \times 1}{}$

$$= \frac{1700 \times N \times 90 \times 650}{1000 \times 60}$$

$= 5207.865$ watts

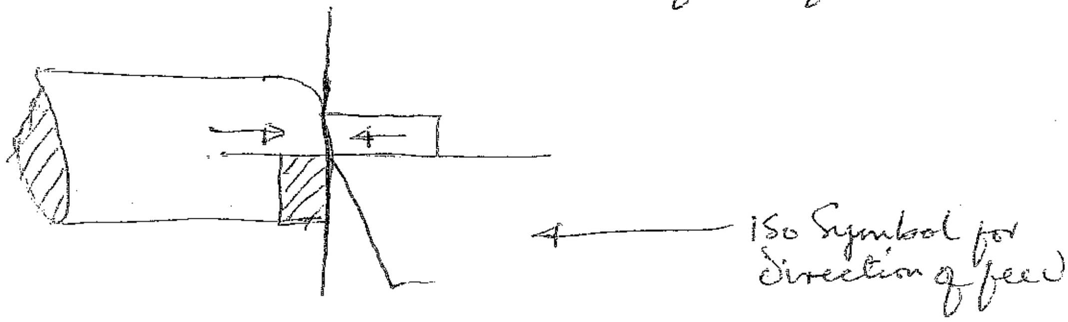
$= 5.208$ Kw

(ii) Power due to feed $= \dfrac{F_1 \times \text{Feed} \times N}{1000 \times 60}$

$$= \frac{500 \times 0.5 \times 550}{1000 \times 60}$$

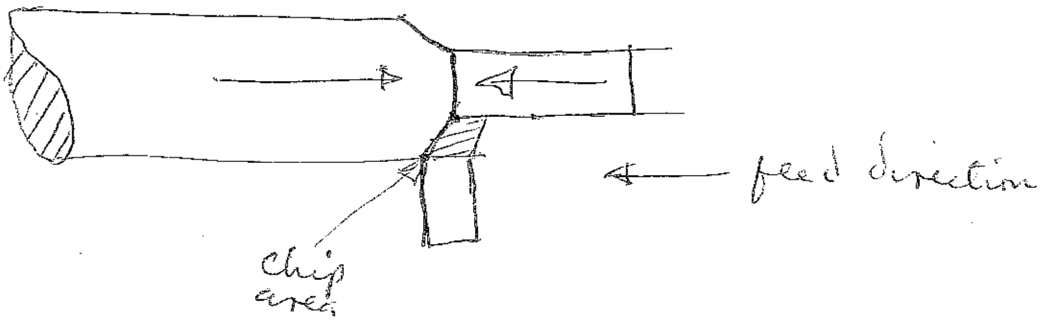
$= 2.7083$ watts.

ORTHOGONAL CUTTING: The figure below shows a typical orthogonal cutting on a lathe machine. In this operation, the cutting edge is perpendicular to the direction of the feed.



OBLIQUE CUTTING

The figure below shows the machining operation where the cutting edge of tool is inclined to the direction of the feed.



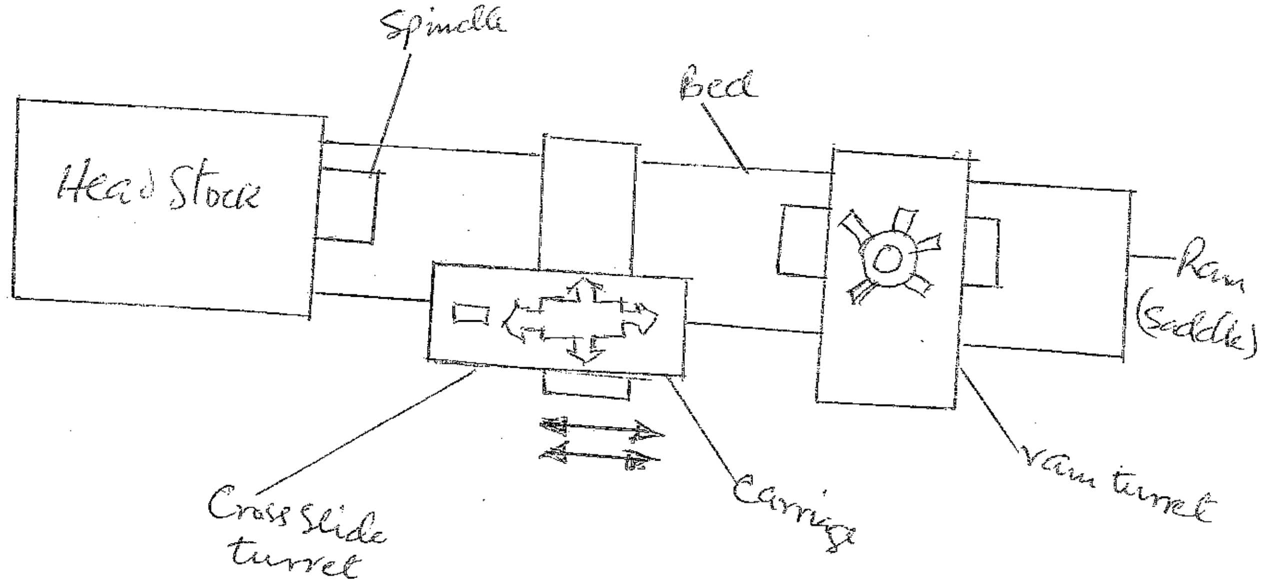
BASIC DIFFERENCE BETWEEN TURRET LATHE AND ENGINE LATHE.

The basic difference between turret lathe and engine lathe are:-

(i) A multi side turret take the place of the tail stock on engine lathe. The turret lathe usually has six sides and is arranged so that tools may be bolted or clamped to each side

The second basic difference is that the turret lathe has a square turret mounted on the cross slide in place of the component rest, on some turret lathe, a fixed tool holder is added at the back end of the cross slide.

There are two types of turret lathe. Viz:- Ram type and Saddle type.



Block diagram showing the basic components of a turret lathe.

Sample Calculation on a turret Lathe Machining

Q. Given the data below, determine the number of pieces above which it will be economical to use turret lathe.

* Time per piece on turret lathe ——— 3½ min
* Turret lathe operator (Main hour) ——— ₦2,000/hr
* Turret lathe over head including interest on Capital ——— ₦3,500/hr
* Set up time ——— 2hrs
* Set up main hour ——— 3,500/hr
* Lathe over head including interest on capital (Engine lathe) —→ ₦2,500/hr.

Solution

From the data given above, the number of pieces above which it will be economical to use turret lathe, we equate the two equations

i.e turret lathe = Engine lathe

$$\left(\frac{3.5}{60}\right)(2000+3500)+\left(\frac{2\times 35000}{N}\right)=\frac{(12)}{60}(3000+2500)$$

$$N = \frac{\left(\frac{3.5}{60}\right)(2000+3500)+\frac{12}{60}(3000+2500)}{2\times 3500}$$

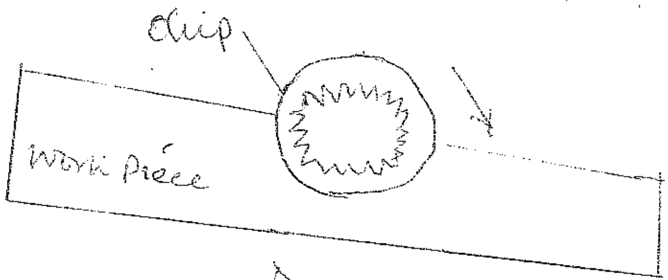
$$= \frac{320.65+1100}{7000}$$

$$= \frac{1420.65}{7000}$$

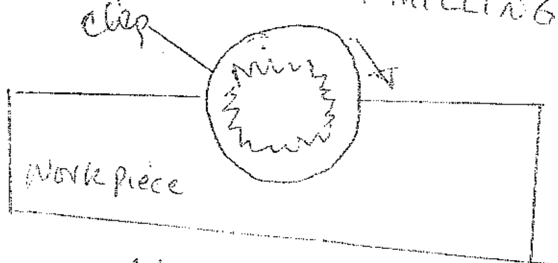
$= N = 0.2$, Piece $= 1$

= even if one piece is required, turret lathe is justified, i.e (economical)

DIAGRAMS SHOWING THE UP CUT AND DOWN CUT MILLING.



DOWN CUT MILLING.



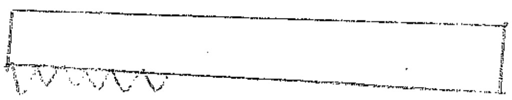
UP CUT MILLING.

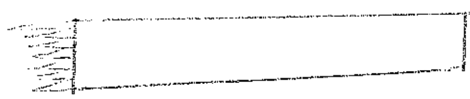
TYPES OF GEARS

(i) ~~Supper~~ Spur gear
(ii) Bevel Gear
(iii) Worm and worm wheel gear.
(iv) Helical gear
(v) Hyperboloid gear.

TYPES OF CHASER.

A chaser is a finishing tool with a number of Screw thread teeth. We have two types;
(i) Inside chaser
(ii) outside chaser.


Inside chaser.


outside chaser.

OPERATIONAL SEQUENCE OF SPUR GEAR MACHINING ON A MILLING MACHINE.

1. Mount the dividing head and it's tail stock on the table on the horizontal milling machine.
2. Check the alignment of head stock and tail stock Centre by Measuring with metre rule, and Squaring the dividing head from the Column face of the Machine of each Centre point.
3. Mount the gear blank on appropriate Mandrel
4. Mount the Mandrel on the dividing head Centre with the large end of the Mandrel toward the head Stock, a Milling Machine dog Should be used for clamping the Mandrel

. Prepare the dividing for (indexing) according to the number of gear teeth required.
. Mount the appropriate gear cutter on the milling arbor and centre it over the gear blanks.
. Set the cutter to proper depth.
. ~~Select the cutter to proper depth~~
. Select appropriate cutting speed and feed.
. take the trial cut from the tail stock end of the blank.

BORING OPERATION.

boring operation is the process of enlarging and turning a hole which is done by a boring machine using a boring tool. they use the working principle of a boring operation on a centre lathe. These machines are designed for machining large and heavy work piece in batch and mass production.

TYPES OF SCREW THREADS.

* Square thread
* Acme thread
* Buttress thread.

TYPES OF ELECTRICAL DISCHARGE MACHINING. (EDM)

There are ~~two~~ three types of electrical discharge machining (EDM): The RAM type EDM, ~~and~~ WIRE type (EDM) and the Electrochemical M/c (ECM).

(i) RAM (EDM): This is a process where by a shaped electrode is made from graphite or copper, the electrode is separated by a non-conductive liquid and maintain at a closer distance (above 0.001). A high DC voltage is pulsed to the electrode and jumps to the conductive work piece and generate a cavity in the versed shape of the electrode or through a hole in the case of plain electrode to permit machining shapes to tight accuracies without the internal stress conventional machining often generates. This is also known as die sinker or siker electrical machining.

WIRE EDM: This is similar to RAM EDM except that small diameter copper or brass wire is used a travelling electrode. The process is usually used in conjunction with computer numeric control (CNC) and will only work when a part is to be cut completely through a common analogy to describe wire electrical discharge machining as an ultra precise electrical contour sawing operation.

ELECTROCHEMICAL MACHINE (ECM): This also use electrical energy to remove material. An electrolytic cell is created in an electrolyte medium with tool as the cathode and the work piece as the anode. A high amperage, low voltage current is used to dissolve the metal and remove it from the work piece which must be electrically conductive. It is essentially a deplating process that utilizes the principles of electrolysis, the ECM tool is positioned very close to the workpiece and a low voltage, high amperage DC current is passed between the two via an electrolyte ~~solutions~~ material removed from the work piece and following electrolyte solutions washes the ions away. These ions from